

Sensor Configuration

A Seaglider talks to its science sensors in one of three ways. Each needs configuring on the glider **and** on the basestation, and most "sensor failures" in the first days of a mission are really configuration problems on one side or the other.

 Source

Paraphrased from 2016–2024 correspondence between Seaglider operators, APL-UW IOP, the manufacturer, sensor manufacturers and service providers (sensor integrations, `.cnf` debugging, CTD and optode faults, basestation processing). Firmware behaviour differs between Rev B (66.x) and Rev E (67.x) builds and between manufacturer and APL-UW firmware – see [Electronics](#) and the IOP firmware pages.

Three ways in

Route	What it is	Typical use
Built-in drivers	Drivers compiled into the firmware (e.g. the SBE CT sail)	The CT, and on Rev B some standard optodes and ECO sensors
serdev	A generic serial driver driven by a <code>.cnf</code> text file: power the sensor, (optionally) send a command, parse the reply into columns	Simple sensors that power up and answer: optodes, ECO pucks, PAR
logdev	A logging-device driver, also <code>.cnf</code> -driven, for instruments with their own memory that log internally and hand back files	GPCTD, PAM recorders, AD2CP, UVP6, echosounders, RBR legato (in some integrations)
SciCon	A separate science controller (Rev B builds) that runs its own sensors to its own schedule (<code>scicon.sch</code>)	Multi-sensor payloads on manufacturer-built gliders